Assignment 2

Exploratory Data Analysis and Logistic Regression from Scratch

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Course	CS 7830 - Machine Learning
Dataset	Flu.csv (410 rows, 18 columns)
Implementation	NumPy, pandas, and Matplotlib only; no machine-learning toolbox

Main result: Forward selection retained `Risk` and `HndWshQual`. On the held-out test set, this model had 81.33% accuracy compared with 77.33% for the model using all 12 features. Feature scaling did not change the final predictions, but it reduced the required gradient-descent iterations from 34,416 to 2,384. The data were imbalanced, so accuracy alone gives an incomplete picture: even the selected model detected only 3 of 16 positive test cases.

Method used

I implemented the sigmoid function, batch gradient descent, train-test splitting, feature scaling, L2 regularization, confusion-matrix metrics, ROC AUC, Wald tests, and forward selection from scratch. Missing rows were removed only for variables required by the current analysis. The test set was 20% of the usable data and was stratified manually with seed 42. The classification threshold was 0.50.

For cross-entropy logistic regression,

$$h_\theta(x) = \frac{1}{1+e^{-X\theta}}, \qquad J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log h_\theta(x^{(i)}) + (1-y^{(i)})\log(1-h_\theta(x^{(i)}))\right].$$

The parameters were updated by

$$\theta \leftarrow \theta - \alpha\frac{1}{m}X^T(h_\theta(X)-y).$$

The cost was saved at every iteration so that convergence could be checked directly.

1. Exploratory Data Analysis

1a. Numeric summary

All eight requested numeric variables have 410 observations. Their summary statistics are below.

Variable	Count	Mean	Std. dev.	Min	25%	50%	75%	Max
Vaccin	410	3.3707	1.3664	1	2	3	5	9
HndWshQual	410	3.8854	1.0367	1	3	4	5	5
HndWshFreq	410	3.2268	1.1553	1	2	3	4	5
SociDist	410	2.9098	1.1760	1	2	3	3	5
NoFaceContact	410	2.4561	1.2467	1	2	2	3	9
RespEttiqu	410	4.1024	1.2506	1	3	5	5	9
PersnDist	410	3.4122	1.1290	1	3	3	4	9
HandSanit	410	2.7854	1.5832	1	1	3	4	5

Table 1: Summary of the requested numeric variables.

HandSanit has the largest standard deviation (1.5832), followed by Vaccin (1.3664). HndWshQual has the smallest spread (1.0367). The maximum value is 9 for Vaccin, NoFaceContact, RespEttiqu, and PersnDist, while most of their quartiles are within 1 to 5. I kept the supplied values unchanged rather than assuming that 9 was an error.

1b. Categorical encoding

Yes. I created readable categorical columns without changing the original numeric variables:

- `Female`: 0 = "No", 1 = "Yes".
- `Sick`: 0 = "Not sick", 1 = "Cold-like symptoms", 2 = "Flu-like symptoms".
- `Flu`: 0 = "No flu-like illness", 1 = "Flu-like illness".

In pandas this was done with `Series.map()` and dictionaries. Missing values remained missing instead of being placed into one of the categories.

1c. Categorical summary and distribution

Variable	Count	Unique	Top category	Top frequency	Distribution comment
Female	375	2	Yes	206	Fairly balanced
Sick	374	3	Cold-like symptoms	213	Cold-like is the majority
Flu	374	2	No flu-like illness	296	Strong class imbalance

Table 2: Categorical summaries after conversion.

Female is reasonably balanced: 206 of 375 known values (54.93%) are Yes. Sick is less balanced: 213 of 374 values (56.95%) are Cold-like symptoms. Flu is clearly imbalanced because 296 of 374 known values (79.14%) are No flu-like illness and only 78 (20.86%) are Flu-like illness. This imbalance helps explain why the models can obtain high accuracy while still missing many positive cases.

1d. Redundant features and other observations

The strongest feature-to-feature correlation is between KnowlTrans and KnowlMgmt, $r = 0.6102$. They may carry overlapping information, so including both can add redundancy and make individual coefficients less stable. I would not automatically delete one based only on this value because the correlation is not near 1.0. HndWshQual and HndWshFreq have the next largest positive correlation, $r = 0.3055$, which is only moderate.

Removing a correlated feature can simplify the model and reduce variance, but it can also remove information that is useful for prediction. I therefore kept all 12 variables in the full model and let forward selection test whether a smaller subset was sufficient. Other observations are the missing values in KnowlTrans (1), Sick (36), Flu (36), and Female (35), the strong imbalance in Flu, and the relatively large spread of HandSanit.

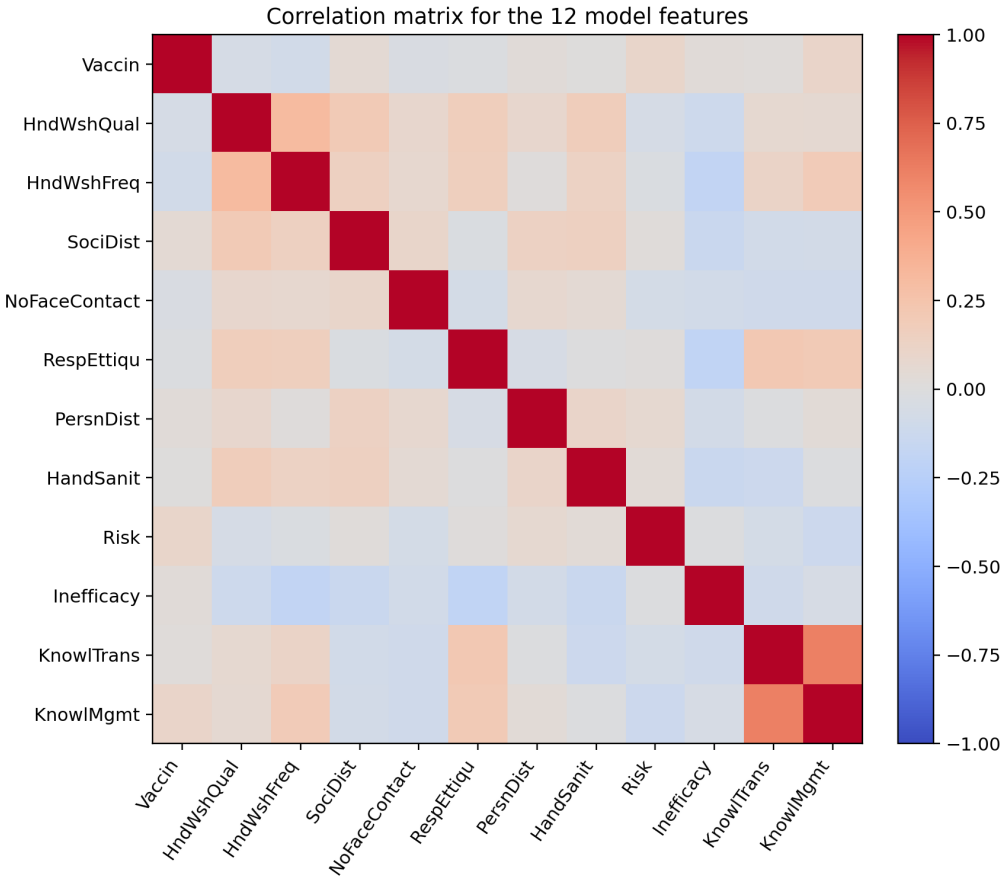


Figure 1: Correlation matrix for the 12 candidate model features.

1e. SociDist and PersnDist

The Pearson correlation between SociDist and PersnDist is $r = 0.1349$. This is a weak positive relationship. Students who reported more social distancing tended to report slightly more personal distancing, but the relationship is small. The two variables are related in meaning, yet they are not redundant in this dataset.

2. Logistic Regression with One Variable

2a. Predicting Flu from Risk

After removing rows with missing Flu, 374 observations remained. The stratified split contained 299 training rows and 75 test rows. Gradient descent produced

$$\theta_0 = -1.3377, \qquad \theta_{Risk} = 1.9518,$$

so the fitted probability was

$$P(Flu = 1 \mid Risk) = \frac{1}{1 + \exp[-(-1.3377 + 1.9518\, Risk)]}.$$

The positive coefficient means that higher Risk is associated with a higher predicted probability of reporting flu-like symptoms.

2b. Test performance

Accuracy	Precision	Recall	Specificity	F1	Balanced acc.	AUC
0.8133	0.6667	0.2500	0.9661	0.3636	0.6081	0.7881

Table 3: Held-out test metrics for the Risk-only model.

The confusion matrix is TN = 57, FP = 2, FN = 12, and TP = 4. The model correctly classifies most students without flu-like symptoms, but it detects only 4 of the 16 positive cases. The 81.33% accuracy is therefore partly caused by the large negative class. The AUC of 0.7881 shows useful ranking ability even though recall at the fixed 0.50 threshold is low.

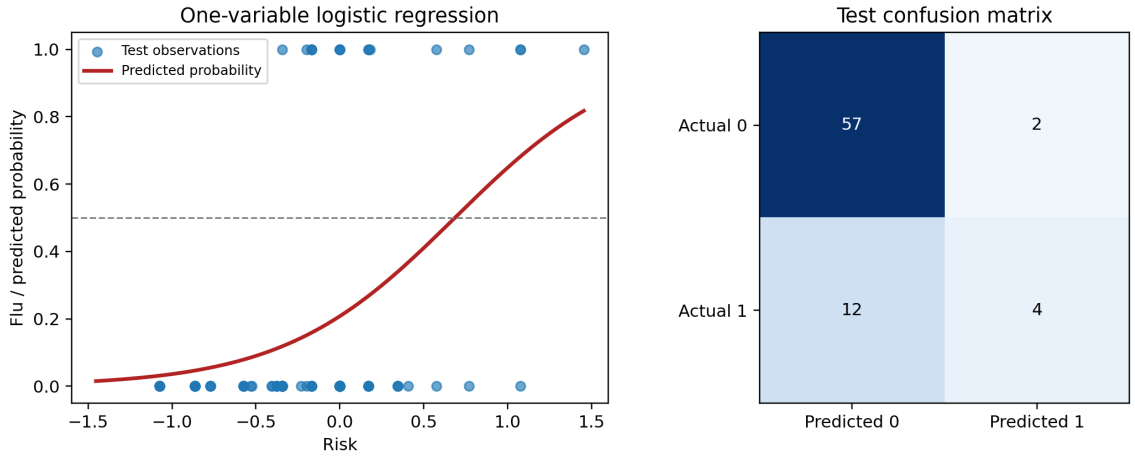


Figure 2: Risk-only predicted probabilities and the held-out confusion matrix.

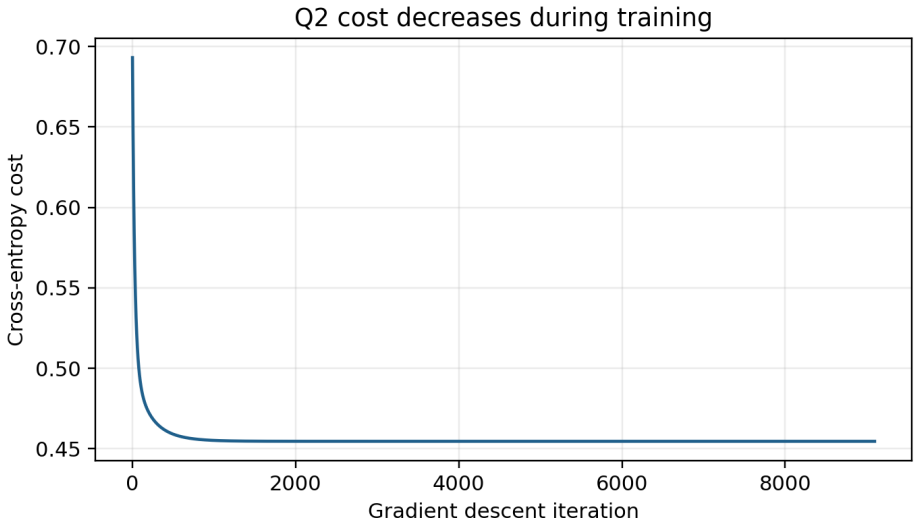


Figure 3: The cross-entropy cost decreases from 0.6931 to 0.4541 during Q2 training.

3. Logistic Regression with Multiple Variables

3a. Full model with 12 features

The full model used the 12 required input variables. One row also had missing KnowlTrans, so 373 complete observations remained. The split contained 298 training rows and 75 test rows. Its test confusion matrix was TN = 57, FP = 2, FN = 15, and TP = 1. The model had 77.33% accuracy, 0.3333 precision, 0.0625 recall, 0.1053 F1, and 0.7733 AUC. It ranked observations reasonably by AUC but almost never crossed the 0.50 threshold for positive cases.

3b. Forward selection

At every step, I fitted each possible candidate model on the training data, calculated the candidate coefficient's Wald p-value from the observed information matrix, and added the variable with the smallest p-value if it was below 0.05. The process stopped after the third check because the best remaining candidate was not significant.

Step	Best candidate	Wald p-value	z-value	Decision
1	Risk	3.3×10^{-8}	5.5253	Added
2	HndWshQual	0.011389	-2.5305	Added
3	PersnDist	0.097089	1.6591	Not added; stop

Table 4: Forward-selection result using the training data only.

The final subset was `Risk` and `HndWshQual`. Risk is reasonable because it directly measures perceived risk of contracting influenza. HndWshQual may add information about prevention behavior that is not completely represented by Risk. Its fitted coefficient was negative after controlling for Risk, meaning higher reported hand-washing quality was associated with a lower fitted probability of Flu in this sample.

This is an association and should not be interpreted as proof of causation.

3c. Full model compared with selected model

Model	Acc.	Prec.	Recall	Spec.	F1	Bal. acc.	AUC	TN	FP	FN	TP
All 12 features	0.7733	0.3333	0.0625	0.9661	0.1053	0.5143	0.7733	57	2	15	1
Risk + HndWshQual	0.8133	0.7500	0.1875	0.9831	0.3000	0.5853	0.7341	58	1	13	3

Table 5: Held-out test comparison for Q3.

The selected model performs better on accuracy, precision, recall, F1, and balanced accuracy. The full model has the higher AUC (0.7733 compared with 0.7341), but its 0.50-threshold classifications are much worse. I kept the selected two-feature model for Q4 because it gave the better classification performance and was simpler.

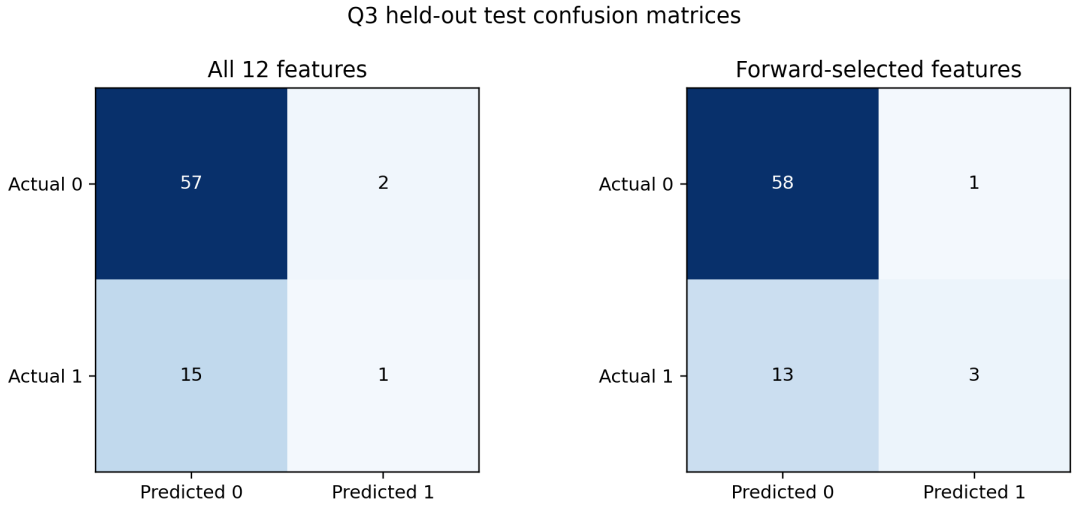


Figure 4: Held-out confusion matrices for the full and selected Q3 models.

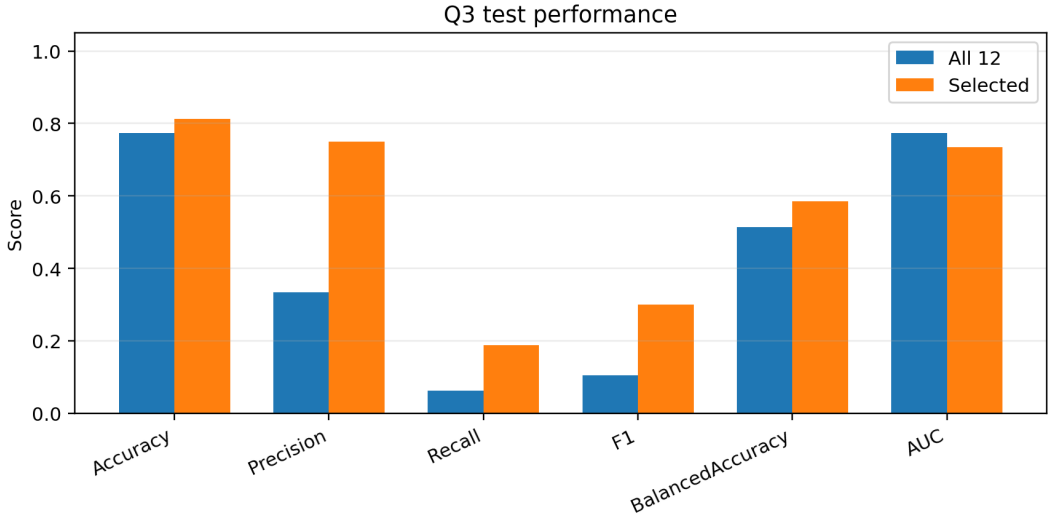


Figure 5: The selected model improves threshold-based metrics, while the full model has higher AUC.

4. Regularization and Cost Function

4a. L2 regularization and feature scaling

I used an internal 25% validation split of the training data to compare $\lambda = 0.01, 0.1, 1$, and 10. The lowest validation log loss occurred at $\lambda = 1$, so the test set was not used to select the regularization strength. L2 regularization was applied to feature coefficients but not the intercept.

Configuration	Iterations	Accuracy	Precision	Recall	F1	Bal. acc.	AUC	Log loss
Baseline	34,416	0.8133	0.7500	0.1875	0.3000	0.5853	0.7341	0.4515
L2, $\lambda = 1$	32,391	0.8133	0.7500	0.1875	0.3000	0.5853	0.7394	0.4522
Feature scaling	2,384	0.8133	0.7500	0.1875	0.3000	0.5853	0.7341	0.4515
Scaling + L2	2,268	0.8133	0.7500	0.1875	0.3000	0.5853	0.7341	0.4516

Table 6: Regularization and scaling experiments on the selected Q3 model.

Regularization did not improve the classification metrics. It slightly increased AUC in the unscaled model, from 0.7341 to 0.7394, but test log loss became slightly worse and the confusion matrix did not change. Feature scaling also did not change the fitted classifications or test metrics. Its clear benefit was computational: convergence required about 93% fewer iterations. I therefore kept the scaled, unregularized cross-entropy model as the Q4a result.

4b. Mean squared error cost

For logistic output $h_\theta(x)$, the required MSE cost is

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right)^2 .$$

Using the chain rule, the derivative used for each parameter was

$$\frac{\partial J}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^{m} \left(h_\theta(x^{(i)}) - y^{(i)} \right) h_\theta(x^{(i)}) \left(1 - h_\theta(x^{(i)}) \right) x_j^{(i)}.$$

Because this objective is non-convex when combined with the sigmoid, I ran it from five random initial parameter vectors. All five runs reached the same final cost of 0.070939 and the same test confusion matrix, so I did not observe different local-minimum solutions in these runs.

Cost	Accuracy	Precision	Recall	F1	Bal. acc.	AUC	TN/FP/FN/TP
Cross-entropy	0.8133	0.7500	0.1875	0.3000	0.5853	0.7341	58/1/13/3
MSE	0.8133	0.7500	0.1875	0.3000	0.5853	0.7267	58/1/13/3

Table 7: Held-out comparison of the two cost functions.

The two objectives gave the same class predictions at the 0.50 threshold, but they did not give exactly the same parameter values or probability ranking. MSE had a slightly lower AUC and a slightly worse cross-entropy log loss. Cross-entropy remains the more natural objective for binary logistic regression because it directly matches the Bernoulli likelihood and has a convex objective for this linear model.

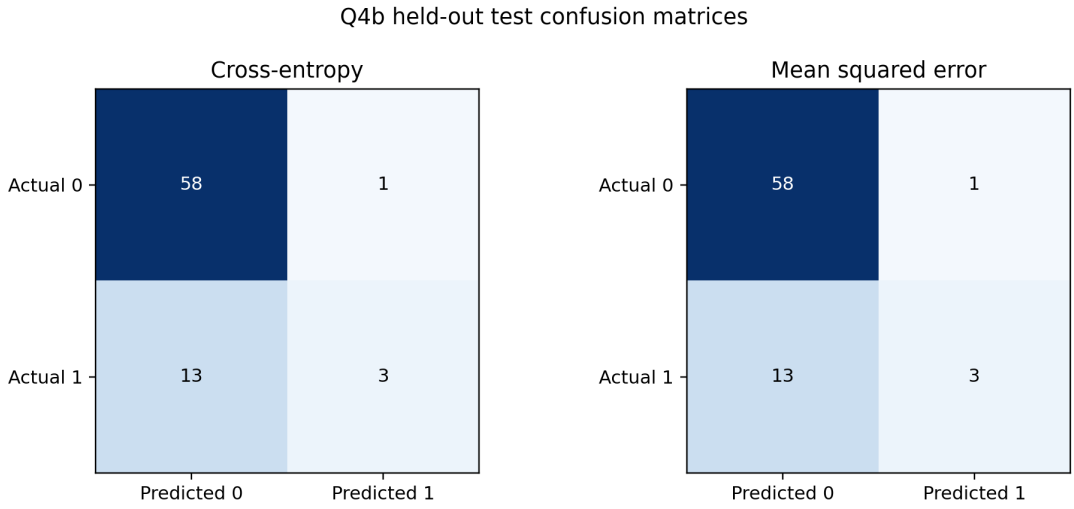


Figure 6: Cross-entropy and MSE produced the same held-out class predictions.

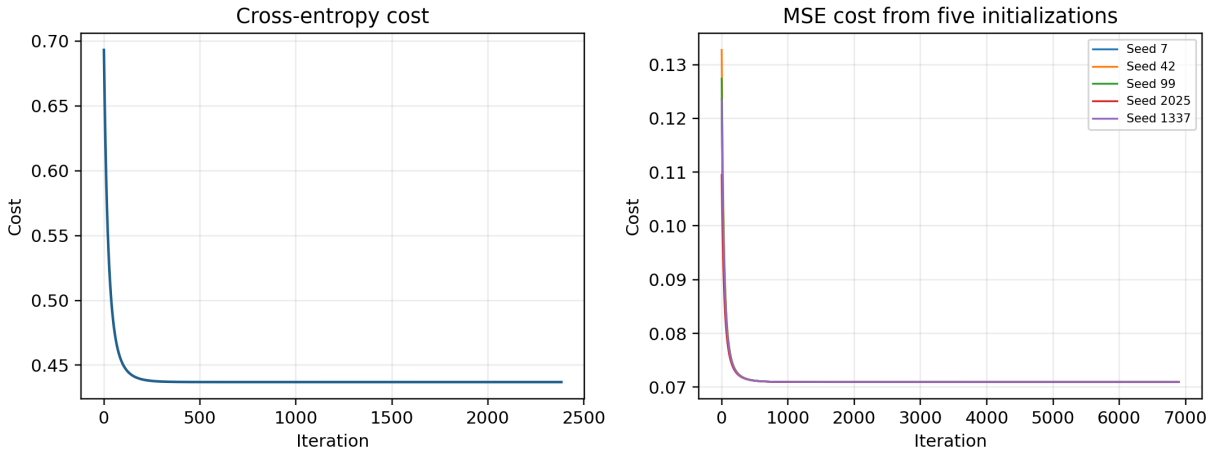


Figure 7: Both objectives decrease. The five MSE initializations converge to the same observed solution.

Final observations

The selected two-feature model is the best threshold-based model in this experiment, but its recall is still only 18.75%. The main limitation is the imbalanced target: only 16 of the 75 test observations are positive. A different threshold could raise recall, but it would also increase false positives. I kept the required 0.50 threshold so every model was compared under the same rule. The results describe this dataset and split; they do not establish causal effects of risk perception or hand-washing behavior.